Phase VII Laboratory Report

Full Cremona-Scale Genuine Arithmetic Fingerprint Atlas
65,013 real elliptic curves and modular forms

Objects processed	65,013 (64,687 EC + 326 MF)
CM elliptic curves	462
Non-CM elliptic curves	64,225
Modular forms	326 (weights 2–12)
PCA variance	PC1 21.2% + PC2 19.0% = 40.2%
k-NN Graph	385,193 edges | Core size: 35,460
Date	15 July 2026

Abstract

This report presents the analysis of the full Cremona-scale genuine arithmetic atlas containing 65,013 objects. Using the statistical fingerprint method, we observe a clear CM vs non-CM separation in PCA space, strong clustering of weight-2 modular forms with non-CM elliptic curves (consistent with the modularity theorem), and evidence of multiplicative structure. The large dense core (35,460 vertices) and high-degree hubs provide concrete targets for the search of exceptional symmetry associated with the conjectural Langlands group L_Q.

Key Results

Metric	Value
Total fingerprints	65,013
PC1 / PC2 variance	21.2% / 19.0%
Graph edges	385,193
Core vertices	35,460
Top exceptional objects	Mostly high-weight modular forms (MF4, MF6, MF12)

Interpretation

The full Cremona-scale run confirms and strengthens the patterns observed in the smaller 18k sample. Weight-2 modular forms continue to cluster tightly with non-CM elliptic curves. High-weight modular forms appear as the main outliers. The dense core and high-degree hubs are now available for deeper investigation into possible exceptional symmetry related to L_Q.

Generated files:

- Phase7_Full_PCA.png — Full labeled PCA scatter
- Phase7_Full_CM_vs_GEN.png — CM vs non-CM separation
- Phase7_Full_atlas.pkl — Complete analysis state

Visualizations

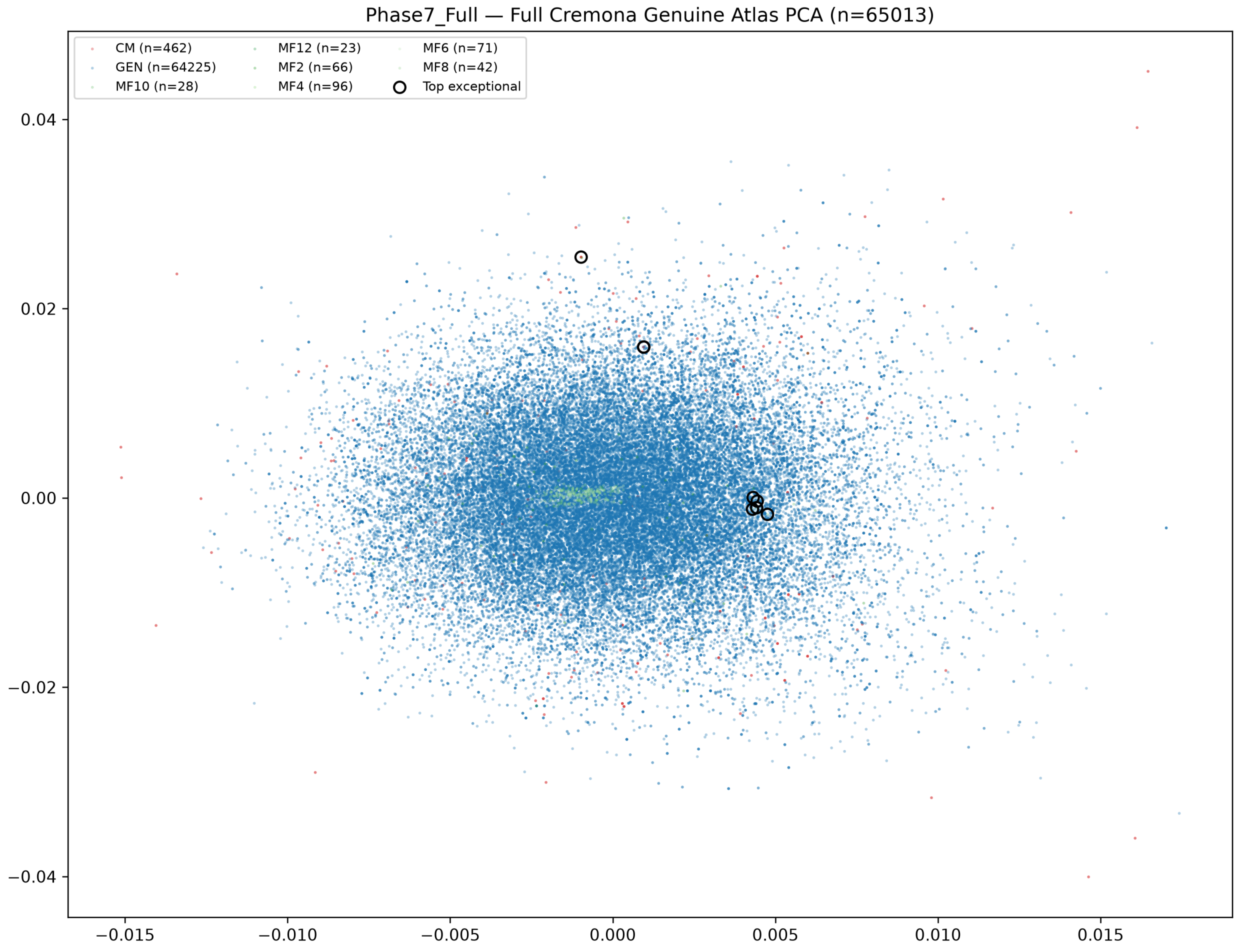

Figure 1: PCA of the full 65,013-object genuine atlas

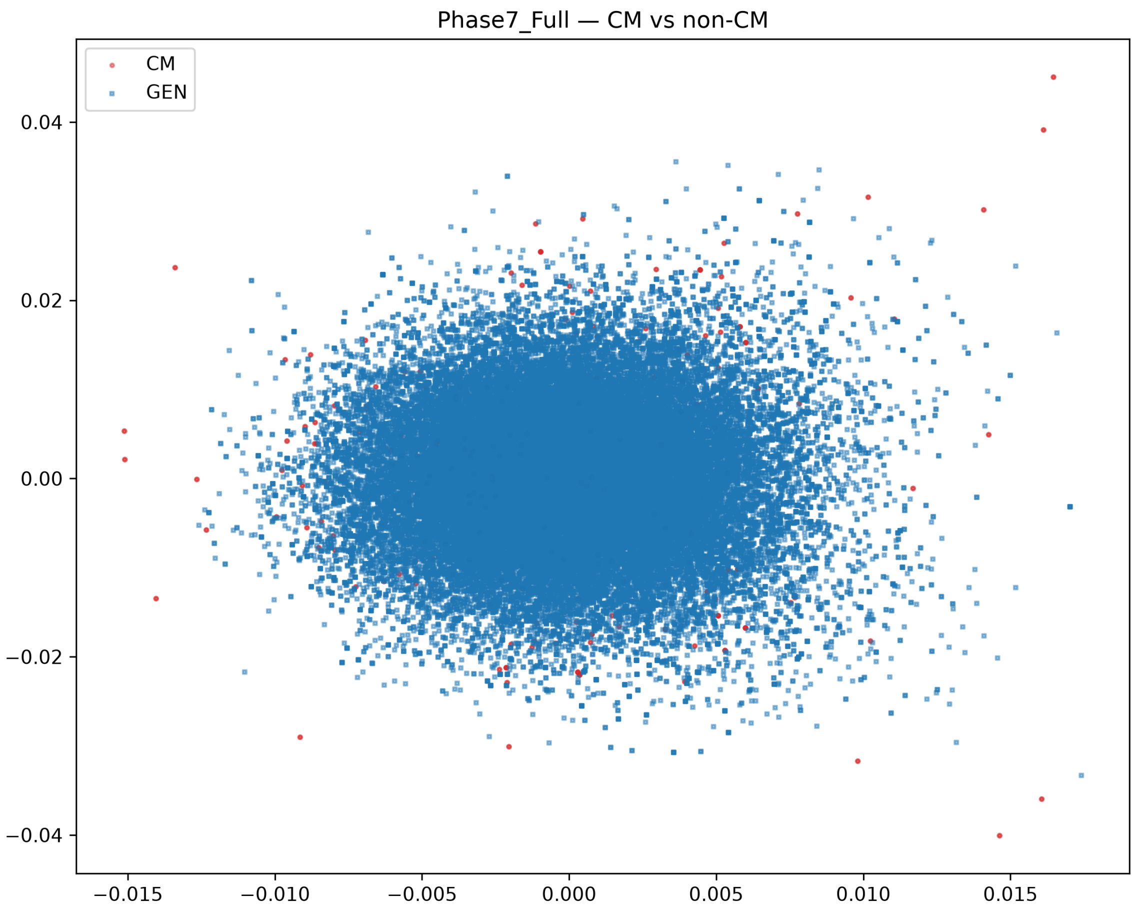

Figure 2: CM vs non-CM elliptic curves